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Comparative Analysis of 3D Reconstruction Results Using MVS and NeRF: Insights from PCA

Maharani M^{1*}, Arif RS², Pradana AR¹, Ayudhia KC¹, Afifah L³, Ikram NM²

¹ Geomatics Engineering Department, Universitas Pembangunan Nasional "Veteran" Yogyakarta, Yogyakarta, Indonesia

² Geodetic Engineering Department, Universitas Gadjah Mada, Yogyakarta, Indonesia

³ Land Monitoring Department, Berau Coal, Berau, Indonesia

*E-mail: monica.maharani@upnyk.ac.id

Abstract. This study examines 3D modeling techniques, emphasizing the advantages of Neural Radiance Field (NeRF) over Multiview Stereo (MVS) in reconstructing accurate models. By employing Principal Component Analysis (PCA), we compare point clouds from both methods to evaluate their quality and distinguish true representations from noise artifacts in practical applications. This approach allows for a detailed assessment of reconstruction quality, highlighting how various factors such as lighting conditions, surface features, and material properties impact the accuracy and density of the resulting 3D models. While NeRF sometimes exhibits a higher point density, MVS demonstrates superior performance, particularly when dealing with homogeneous textures, yielding denser point clouds and more accurate representations. The analysis shows that MVS excels in data density, feature extraction, and noise reduction, resulting in consistently cleaner models. In contrast, NeRF, despite its high data density, is adversely affected by significant noise and outliers, which obscure object details. Both methods achieve satisfactory levels of object completeness; however, MVS outperforms NeRF in detail sharpness, surface smoothness, and overall clarity. This comparison underscores the critical influence of texture and surface characteristics on the effectiveness of 3D reconstruction techniques, affirming MVS's advantages in producing reliable and accurate representations of 3D objects.

1. Introduction

3D models have become a critical necessity across various fields in the contemporary era. There are several methods for 3D modeling recognized by the general public; however, within the engineering domain, 3D modeling can be accomplished not only through laser scanning techniques but also via close-range photogrammetry [1], [2], [3]. Close-range photogrammetry, which employs non-metric cameras, falls under the category of low-cost photogrammetry and has been widely utilized for 3D modeling in many applications [4], [5]. Although it remains an alternative for 3D modeling, close-range photogrammetry is still subject to limitations, as the quality of the resulting 3D models is significantly influenced by the conditions under which the images are captured.

Multiview Stereo (MVS) has emerged as a pivotal technology in recent times for addressing challenges in 3D reconstruction from photogrammetric data. Numerous studies have demonstrated that MVS can perform exceptionally well during the 3D reconstruction process [6],



[7], [8], [9], [10]. However, it is important to note that MVS may encounter difficulties, particularly when dealing with textureless objects [11], [12], [13]. Consequently, various algorithms have been developed to mitigate these reconstruction challenges.

Deep learning has gained significant attention in light of the rapid developments in 3D acquisition technologies, especially within the geospatial field [14]. It has proven instrumental in advancing various 3D modeling projects, resulting in models closely mimicking their real-world counterparts [15], [16]. Recently, new techniques have emerged for generating 3D models from photogrammetry that claim to outperform the reconstruction capabilities of Multiview Stereo (MVS). One notable method is the Neural Radiance Field (NeRF), which is said to create highly accurate 3D models using only photographic data [17].

NeRF offers significant benefits in constructing 3D models from photogrammetric data, primarily due to its use of Neural Surface Reconstruction (NeUS) [17]. This approach effectively reconstructs 3D forms by analyzing edges detected in images and estimating the camera's position. Additionally, NeRF has been tested on basic objects, including small statues and staircases, and has been shown to exceed the performance of traditional methods like MVS and Structure from Motion (SfM). It demonstrates the ability to automatically reconstruct objects and generate shapes that closely resemble the originals, even with a limited number of input images [18], [19], [20], [21], [22].

Numerous studies assert that Neural Radiance Field (NeRF) is a superior method for 3D object reconstruction, often surpassing the performance of Multiview Stereo (MVS). This investigation seeks to provide an alternative perspective by evaluating the quality of point clouds generated by NeRF compared to those produced by MVS, employing Principal Component Analysis (PCA) as the analytical approach. The utilization of PCA in this comparative analysis serves to identify and reduce the dimensionality of the point cloud data, thereby facilitating a clearer (less noise) and more efficient examination of the geometrical structures within the datasets [23]. Assessing the quality of point clouds is paramount, as it enables the discernment of whether the reconstructed points accurately represent the original object or are merely artefacts of noise that may have been captured during the reconstruction process. Through this analysis, we aim to enhance our understanding of the accuracy and reliability of the employed methodologies and their practical implications in real-world applications.

2. Data and Methods

This study employs 3D reconstruction techniques, specifically Multi-View Stereo (MVS) and Neural Radiance Fields (NeRF), to create detailed models from multiple images of objects. The dataset consists of high-resolution images captured through videogrammetry for Object 1 and photogrammetry for Object 2, serving as input for both methods and facilitating comparative analysis of their performance regarding point cloud density and visual quality. Additionally, Principal Component Analysis (PCA) is utilized to evaluate the distribution and clarity of features within the generated models, enhancing the overall assessment.

2.1 Videogrammetry Data

The analysis in this study utilizes two objects with distinct characteristics. Object 1 is a small statue measuring 60 x 80 x 20 centimeters, made of stone. The statue features a relatively uniform grey colour and exhibits a low texture. It depicts a 3D solid representation of a sitting elephant. Although there are no other objects present, the statue includes several intricate details, such as the hands holding the trunk, carved fingers, and other subtle contours, as illustrated in Figure 1(a). The videogrammetry method was employed to capture data on the statue. The statue was recorded using a rotating camera that encircled the object, ensuring all angles were covered in the

video. This single recording session captured all the intricate details of the statue within its frames. The video has been extracted to 90%, resulting in a distribution closely aligned with Figure 2(a).

2.2 Photogrammetry Data

Object 2 is an imposing statue measuring 180 x 180 x 300 centimetres, characterized by its intricate form Figure 1(b). The base consists of a pentagonal concrete structure adorned with a rough ceramic finish, while atop it rests a series of five figurative sculptures, each exhibiting distinct shapes and spatial arrangements. The concrete base features a relatively uniform grey hue complemented by a textured surface, whereas the statues are rendered in a striking bronze colour. To acquire detailed data on the statue, the photogrammetry method was employed. This involved a rotating UAV camera that circumnavigated the object, ensuring comprehensive coverage of all angles in the captured photographs Figure 2(b).



Figure 1. (a) Object 1 – Ganeca Statue (b) Object 2 – Dharma Eva Hato Hanti Statue.

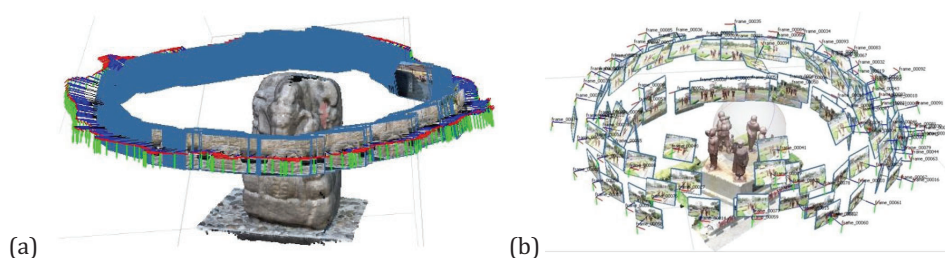


Figure 2. Camera Configuration (a) Object 1 (b) Object 2

2.3 Data Processing

All images were subsequently processed to achieve pose estimation for each photograph. The SfM-MVS method was employed to determine camera orientation estimation parameters, convert photographs into a sparse point cloud, and produce a dense point cloud. Following this, a 3D mesh model was created for each object. The software utilized for this process was Agisoft Metashape version 2.1.0, which was configured with high-quality settings for alignment and dense point cloud construction.

The NeRF method was then applied, utilizing the camera orientation estimation parameters obtained from the SfM-MVS method. This data served as input for training a Multi-Layer Perceptron, facilitating the synthesis of new views and the creation of a volumetric model. The Nerfstudio library, developed by Tancik et al. [24], was employed for this purpose. By default, this

application reduces image dimensions through a scaling factor to expedite the training process (downscaling) [25]. However, in this study, various strategies were implemented to ensure that the data remained undownscaled, thereby preserving the integrity of the data quality.

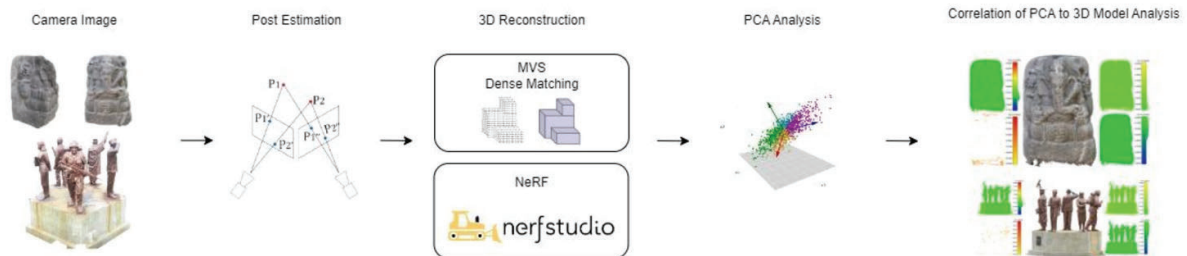


Figure 3. Research Flow

The method described involves a comprehensive approach to 3D reconstruction using Neural Radiance Fields (NeRF) and Multi-View Stereo (MVS) techniques, which generate point cloud data that represents the spatial distribution of a 3D object. This process is sensitive to various influencing factors:

- **Lighting Condition:** Variations in illumination can affect how surface features are captured, leading to differences in the resulting point cloud.
- **Surface Features:** The texture and geometry of the object being reconstructed can impact the accuracy and density of the point cloud data.
- **Material Properties:** Different materials reflect light in distinct ways, influencing how well the object is captured in the reconstruction process.

As a result of these factors, the point cloud data often contains varying noise levels, which can affect the quality and reliability of the reconstruction [26].

To analyze the consistency of this point cloud data, Principal Component Analysis (PCA) is utilized. PCA is a statistical technique that transforms the data into a set of values (principal components) that capture the most variance, allowing for dimensionality reduction while preserving the essential patterns. It fundamentally assumes a Gaussian distribution of the data, which allows for standardized evaluation. In this study, the PCA results are represented through standard deviation calculations, providing insights into the consistency and reliability of the point clouds generated by NeRF and MVS. The point clouds are then categorized into three distinct clusters based on their PCA scores, with defined ranges of (0.4 – 0.6), (0.6 – 0.8), and (0.8 – 1.0). This segmentation helps to identify areas of varying consistency, with higher ranges indicating more reliable data and lower ranges suggesting potential noise or inconsistency in the point cloud representation. Overall, this method provides a structured way to assess the quality and reliability of 3D reconstructions of their capturing conditions.

3. Results and Discussion

The analysis of visual characteristics in 3D reconstruction aims to assess both the visual quality of the generated 3D model and the completeness of the reconstruction. Key components of this analysis include evaluating the quality and quantity of point cloud density, conducting Principal Component Analysis (PCA) on the point cloud, and analyzing the resulting 3D model. Both Multi-View Stereo (MVS) and Neural Radiance Fields (NeRF) methodologies will be explored in depth.

3.1 Point Cloud Density of the 3D Reconstruction Results

The density of the point cloud in 3D reconstruction is a critical metric that reflects the quality and completeness of the generated model. It indicates how densely packed the points are in the reconstructed space, influencing the overall fidelity of the representation. A higher point cloud density typically correlates with better detail and accuracy in capturing the features of the object or scene.

For the object 1, the MVS method generated 1,415,494 points, whereas the NeRF method produced 1,599,189 points. The density levels achieved by the MVS and NeRF methods for object 1 indicate that the reconstruction results from NeRF are significantly more abundant than those obtained from MVS. This difference adversely affects the visual quality of the reconstructed models. While the color patterns of both models are consistent, the MVS model exhibits big blank spots (Figure 4).



Figure 4. Comparative Analysis of Point Cloud Visual Results for Object 1

Although the MVS result contains a blank spot, the differences between the two results are not prominently evident from the front view. However, when exploring the top view, the significant blank spot reveals that MVS failed to reconstruct certain object parts. It is indicated that the MVS method's limitations arise from the object's characteristics, which exhibit a homogeneous colour and low texture. This observation aligns with [27], who suggest that MVS reconstruction results are influenced by the texture type of the object's surface. In contrast, NeRF demonstrates superior results compared to MVS, as evidenced by the denser point cloud. This superiority is attributed to NeRF's reliance on volume rendering, which is not dependent on the object's texture type [27].

In contrast to Object 1, for Object 2, MVS produced 1,560,670 dense points, while NeRF generated 1,179,739 points. In the case of Object 2, MVS surpasses NeRF in terms of point cloud density richness, and the visual appearance is markedly distinct. MVS presents what appears to be a clear 3D model, even in the absence of mesh formation (Figure 5).

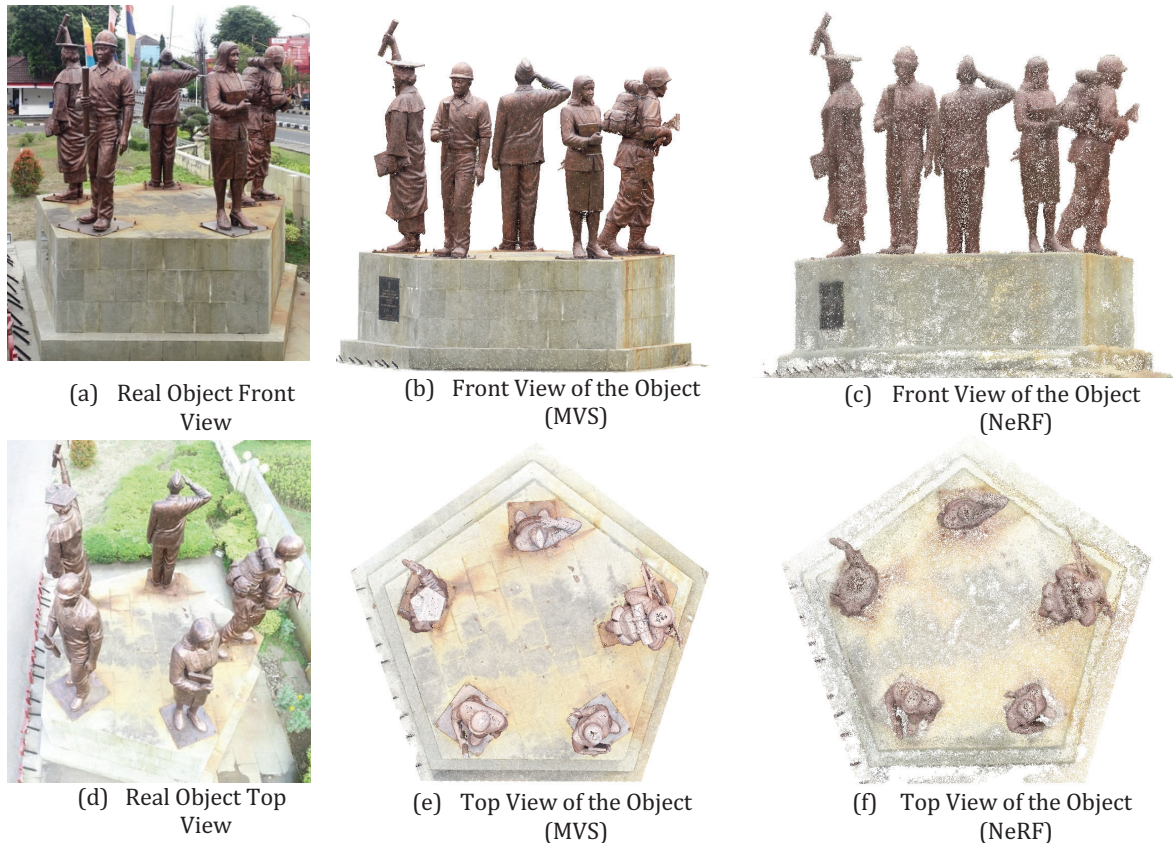


Figure 5. Comparative Analysis of Point Cloud Visual Results for Object 2

In Object 2, the MVS results are notably superior to those of NeRF due to the abundance of the generated point cloud. This richness contributes to a more realistic appearance that closely resembles the actual scene in the field, in contrast to the point cloud results produced by NeRF. When viewed from above, the differences are evident, as the outlines of the ceramic objects are more distinctly defined using the MVS method, reflecting a more accurate depiction of on-site conditions compared to NeRF. This disparity arises from the significantly lower density of the point cloud generated by NeRF relative to MVS. In Object 2, NeRF's point cloud representation is less effective, as indicated by its reliance on volume rendering. For collections of solid objects with distinct features that are not fused, volume rendering in NeRF has yet to surpass the performance of MVS. In contrast, the MVS method excels due to the heterogeneous surface characteristics of the objects in Scheme 2, which possess abundant textures, allowing for the generation of rich features [27].

3.2 Principal Component Analysis (PCA) for Density of the 3D Reconstruction Results

3.2.1 PCA for Density of Object 1

The PCA distribution results obtained from the front and top view reconstruction using the MVS method are illustrated in Figure 6. The findings indicate that the data distribution across the three defined ranges demonstrates excellent density, with the highest concentration occurring within the PCA range of 0.4 to 0.6, while the lowest concentration is observed in the PCA range of 0.8 to 1.0. Additionally, the PCA analysis suggests that feature extraction from MVS exhibits high quality, as evidenced by the substantial density of the data. Notably, this method shows low noise detection, characterized by a minimal number of outlier data points. The effective data distribution further contributes to the overall reduction of noise.

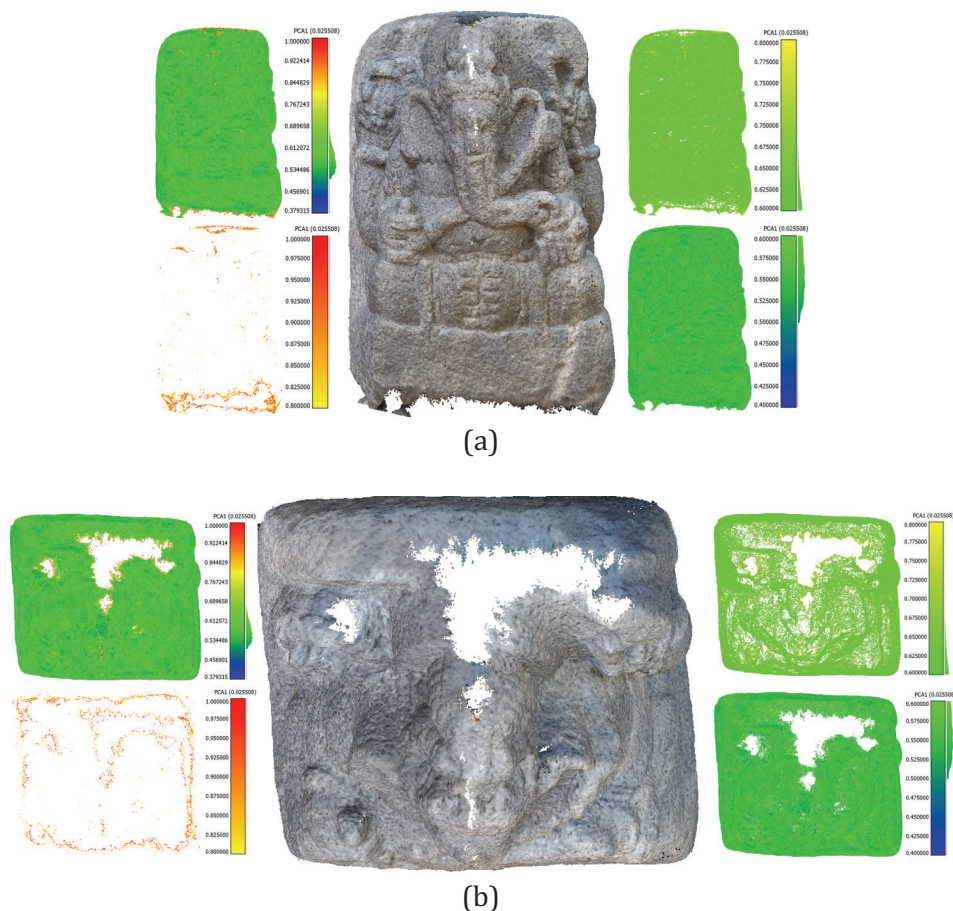


Figure 6. PCA Analysis of 3D Object 1 Using MVS (a. Front View; b. Top View)

In contrast with MVS, the PCA distribution results obtained from the front and top view reconstruction using the NeRF method can be seen in Figure 7. The results indicate that the data distribution across the three defined ranges demonstrates excellent density, although the highest concentration of data occurs within the PCA range of 0.4 to 0.6, while the lowest concentration is observed in the PCA range of 0.8 to 1.0. The PCA analysis also suggests that the feature extraction from NeRF exhibits high quality, as indicated by the data's substantial density. However, in the reconstruction results of NeRF across each data range, there is a noticeable presence of noise. This is evident in the noise detection, characterized by the number of outlier data points deviating from

the majority range, as indicated by the colour differences in the distribution from the predominant data cluster.



Figure 7. PCA Analysis of 3D Object 1 Using NeRF (a. Front View; b. Top View)

3.2.2 PCA for Density of Object 2

In Object 2, results similar to those of Object 1 are evident. The PCA distribution results from the front view reconstruction using the MVS method are shown in Figure 8, while the top view is illustrated in Figure 9. The analysis indicates that the data distribution across the three specified ranges exhibits excellent density, with the highest concentration occurring within the PCA range of 0.4 to 0.6 and the lowest concentration observed in the PCA range of 0.8 to 1.0. Additionally, the PCA analysis suggests that feature extraction via MVS displays high quality, as evidenced by the substantial density of the data. Notably, this method exhibits minimal noise detection, characterized by a limited number of outlier data points. The effective distribution of data further contributes to the overall reduction of noise.



Figure 8. PCA Analysis of 3D Object 2 Using MVS - Front View

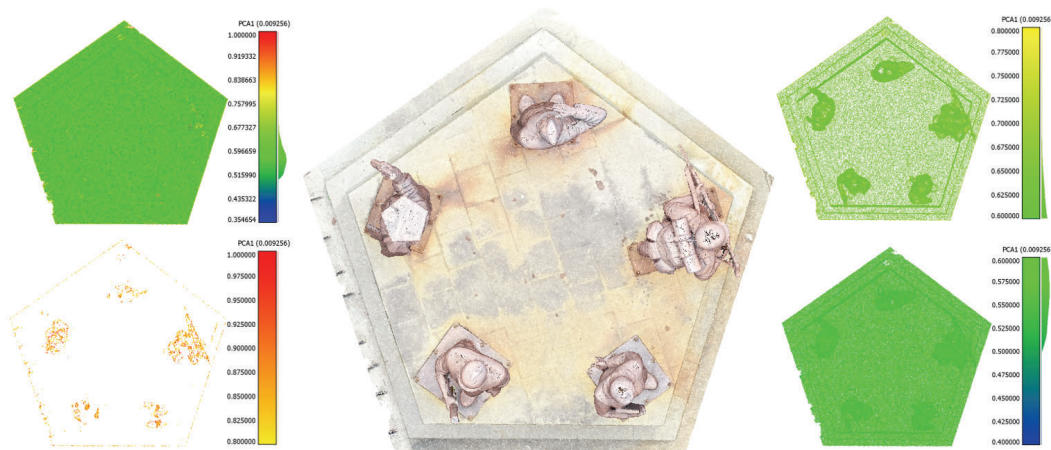


Figure 9. PCA Analysis of 3D Object 2 Using MVS - Top View

Same pattern as Object 1, the PCA distribution results from the front (Figure 10) and top view (Figure 11) reconstructions using the NeRF method show excellent data density. While the highest concentration is found in the PCA range of 0.4 to 0.6, the lowest is in the range of 0.8 to 1.0. The analysis indicates high-quality feature extraction from NeRF but also reveals a significant presence of noise, evidenced by outlier data points deviating from the majority range.



Figure 10. PCA Analysis of 3D Object 2 Using NeRF - Front View

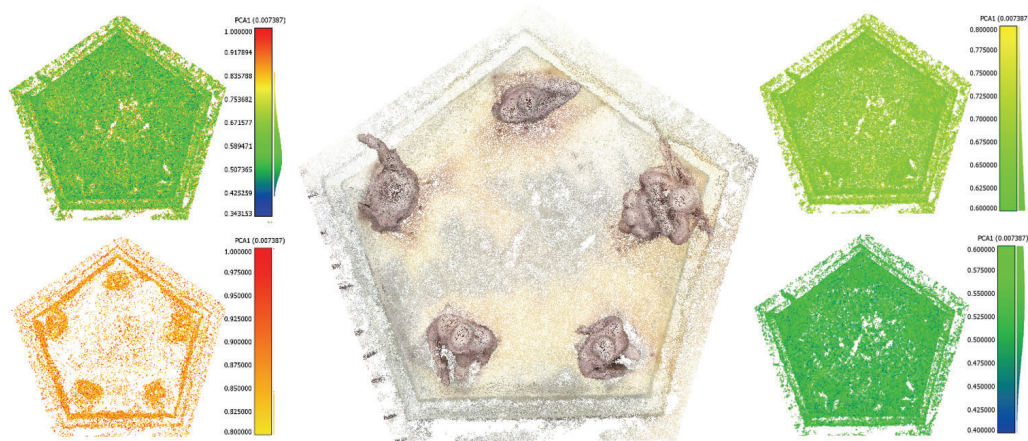


Figure 11. PCA Analysis of 3D Object 2 Using NeRF - Top View

3.3 3D Model Representation

The analysis of the 3D models Object 1 generated by both MVS and NeRF encompasses an evaluation of the visual quality achieved and the level of completeness exhibited in the 3D models. The 3D models generated by MVS and NeRF exhibit a good level of object completeness. This is evident from the comprehensive visual appearance of both objects (Figure 12(b) and Figure 12(d)). Although some blank data points are visible in the solid models (Figure 12(a) and Figure 12(c)), the interpolation in the generated mesh adequately fulfils the completeness of the objects. From the completeness perspective, both the NeRF and MVS models demonstrate a satisfactory level of completeness, as seen in Figure 12(b) and Figure 12(c).



Figure 12. (a) MVS Solid Model; (b) MVS Mesh Model; (c) NeRF Solid Model; (d) NeRF Mesh Model

However, the resulting 3D models from MVS and NeRF show significant differences in detail sharpness, surface smoothness, and clarity of object structure. MVS exhibits superior sharpness, surface smoothness, and clarity compared to NeRF. This can be correlated with the PCA monitoring results that have been conducted.

Similar to Object 1, in Object 2, MVS exhibits enhanced detail, smoother surfaces, and greater clarity than NeRF.



Figure 13. (a) MVS Mesh Model (Front-View); (b) NeRF Mesh Model (Front-View); (c) MVS Mesh Model (Back-View); (d) NeRF Mesh Model (Back-View)

From Object 2, it becomes increasingly evident that MVS exhibits enhanced detail, smoother surfaces, and greater clarity than NeRF. This is evident in Figure 13(a) and Figure 13(c), which showcase the intricate details of the objects produced by the MVS method. In this case, NeRF generates a 3D model filled with noise, obscuring the intended features of the object. Various unnecessary contours are instead depicted in NeRF, which detracts from the overall visualization of the object in Figure 13(b) and Figure 13(d).

Table 1. Comparative Analysis of MVS and NeRF

Parameter	MVS	NeRF
Data Distribution	High Quality	High Quality
Feature Extraction	High Quality	High Quality
Noise Detected	Low	High
Completeness of the 3D Model	Good Quality	Good Quality
Detail Sharpness	Clear and Precise	Vague and Indistinct
Surface Smoothness	Seamless Surface	Rough Surface
Clarity of Object Structure	Detail	Obscure

When the 3D results are analyzed in correlation with PCA analysis, the comparison of the outcomes is in **Table 1**. NeRF demonstrates a commendable quality in data distribution, feature extraction, and completeness of 3D models, similar to MVS. However, NeRF struggles to reconstruct 3D models as effectively as MVS. The primary limitation lies in the noise in the NeRF-generated models, which affects their overall quality.

4. Conclusions

In summary, the analysis of point cloud density reveals significant differences between the MVS and NeRF methods across both objects. For Object 1, NeRF shows a greater abundance of points, yet MVS struggles with certain reconstructions due to the object's homogeneous texture. Conversely, for Object 2, MVS outperforms NeRF with a denser point cloud, resulting in a more accurate and realistic representation. This highlights the importance of texture and surface characteristics in determining the effectiveness of 3D reconstruction methods, as MVS benefits from its ability to handle complex textures, while NeRF's reliance on volume rendering limits its performance in this context.

The PCA distribution analysis reveals distinct differences in data quality and noise levels between the MVS and NeRF methods across both objects. MVS consistently demonstrates excellent data density and effective feature extraction, with minimal noise and outliers, contributing to a high-quality representation of the object structure. In contrast, while NeRF also shows high data density and quality feature extraction, it is marred by a significant presence of noise, indicated by numerous outlier data points. This pattern persists in both Object 1 and Object 2, underscoring MVS's superiority in producing clearer, more reliable 3D reconstructions compared to NeRF. The analysis of 3D models for Object 1 and Object 2 reveals that both MVS and NeRF achieve a satisfactory level of object completeness, as demonstrated by their comprehensive visual appearances.

However, significant differences emerge in detail sharpness, surface smoothness, and clarity, with MVS consistently outperforming NeRF. The intricate details showcased in the MVS-generated models highlight its superiority, while NeRF's output suffers from noise and unnecessary contours that obscure the object features. Despite NeRF's commendable performance in data distribution and feature extraction, its effectiveness is hindered by noise, impacting the overall quality of the reconstructed models. This comparison underscores the advantages of MVS in achieving clearer and more accurate representations of 3D objects.

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